

Your grade: 100%

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Next item →

1. Which of the following statements is true?

1 / 1 point

- ☐ An axiom comprises necessarily multiple quantifiers
- ☒ A proof is a valid argument used to prove the truth of a statement
- ☐ A lemma is equivalent to a specific rule of inference

✔ Nice work

2. Given the statement S: 'For every integer n , there exists a number dividing n':

1 / 1 point

- ☐ S is a law of logic
- ☒ S can be formalised as : $\forall n \in N \exists p \in N p/n$
- ☐ S can be proved only using contrapositive

✔ Nice work

3. How can we build a valid direct proof?

1 / 1 point

- ☐ We use axioms, definitions and theorems, together with rules of inference, to show that if $\neg q$ is true then $\neg p$ must also be true
- ☒ We show that the conditional statement : $p \rightarrow q$ is true
- ☐ We start by assuming that $\neg q$ is true and then prove that $\neg p$ must also be true

✔ Nice work

4. How can we build a valid proof by contrapositive?

1 / 1 point

- ☐ We show that the conditional statement: $p \rightarrow q$ is true
- ☒ We start by assuming that $\neg q$ is true and then prove that $\neg p$ must also be true
- ☐ We use axioms, definitions and theorems, together with rules of inference, to show that if $\neg q$ is true then $\neg p$ must also be true

✔ Nice work

5. How can we build a valid proof by contradiction?

1 / 1 point

- ☒ We start by assuming that $\neg p$ is true and show that $\neg p \rightarrow False$
- ☐ We show that the conditional statement: $\neg q \rightarrow \neg p$ is true
- ☐ We use axioms, definitions and theorems, together with rules of inference, to show that if p is true then q must also be true

✔ Nice work